



Grow with Data
growwithdata.net

Industrial Training Program

AI/LLM Application Builder From Zero to Mastery

A practice-first industrial training in modern AI application engineering — one real system, built session by session, deployed live.

10 Weeks

20 Sessions

40 Contact Hours

Friday & Saturday · **7:30–9:30 PM**

Live · Instructor-led



Prepared for the **Department of Computer Science and Engineering** — as part of the students' industrial training program.

Program overview

This training bridges the gap between academic coursework and industry AI engineering practice. Instead of surveying disconnected topics, participants spend all twenty sessions building **one production-style AI system** — the way a real engineering team would: incrementally, with version control, testing, observability, and a final live deployment. Every session ends with the project visibly more capable than it began, and every version is committed so students can trace exactly what they built and why. The program uses the current industry stack — Python, LangChain & LangGraph, retrieval (RAG), FastAPI, React, LoRA/QLoRA fine-tuning of open-weights models, and AI-assisted development with coding agents (Claude Code).

The training project — “ShopWise”, an AI service desk for an online store

A typical e-commerce shop drowns in support messages: “Where is my order?” · “Can I return this?” · “I was charged twice!” Students build the complete AI system that runs this shop's service operation:

WHO IT SERVES



Customers

A 24/7 support chat that checks real orders & delivery status, explains return and warranty policy from the company's actual documents, and hands over to a human when it should.



Support manager

Nothing risky happens alone: every refund waits for the manager's approve / adjust / reject — from any device, with a full audit trail.



Shop owner

A one-page business brief every morning — sales, refunds, complaints — plus live market and competitor price research.



THE APPLICATION STUDENTS BUILD

The ShopWise web application



Support chat

replies stream live, word by word



Approval screen

refunds reviewed & signed off



Daily report page

the owner's morning brief



THE INTELLIGENCE BEHIND IT

A team of AI agents working together

a supervisor assigns each request to the right specialist



Supervisor agent



Orders & delivery expert

looks up real order data



Policy expert

quotes the exact clause — or declines



Refunds expert

pauses & waits for manager approval



Market analyst

researches the web, cites sources



— WHAT IT DRAWS ON



Store systems

Customers, orders, shipments, products, refunds.



Company documents

Return, shipping & warranty policies — searchable and quotable.



The live web

Competitor prices & market information, with cited sources.

Under the hood — the industry engineering practices students implement to make this real:

- ✓ Tool-using AI agents grounded in real data (never inventing facts)
- ✓ Human-in-the-loop approval workflows that pause and resume
- ✓ Safety engineering: prompt-injection filters, PII redaction, cost limits
- ✓ Full observability — every AI decision traceable and debuggable
- ✓ Fine-tuning a small open-weights model on the shop's own data (LoRA/QLoRA) and evaluating it honestly
- ✓ Document search & retrieval (RAG) with citations and honest refusals
- ✓ Conversation memory per ticket + durable customer profiles
- ✓ Multi-agent orchestration with explicit, auditable workflows
- ✓ Web API + modern web interface + cloud deployment & scheduling
- ✓ AI-assisted engineering: product specs, task breakdown, coding-agent skills, hooks and MCP integrations

Session-by-session plan

Wk	S#	Session focus	The project after this session
1	S1	How AI language models actually work — tokens, context, why they hallucinate; environment setup	Foundations laid; tools installed and verified
	S2	Talking to LLMs — messages, roles, and system prompts that control behavior	First automated support reply, in the company's voice
2	S3	Outputs software can trust — structured data, error handling, cost tracking	Tickets triaged into category, urgency & sentiment
	S4	LangChain: one interface for every AI provider + LangSmith observability	Vendor-independent pipeline, every call traceable
3	S5	Tools & agents — giving the AI “hands” to look things up	Answers from real order, shipping & product data
	S6	Conversation memory & live streaming replies	Multi-turn support chat that remembers the ticket
4	S7	Safety rails — guardrails, PII protection, human approval on risky actions	Refunds now require a manager's sign-off
	S8	Teaching it the company's documents — indexing & semantic search (RAG part 1)	Policy documents searchable by meaning
5	S9	Grounded answers — citations and honest refusals (RAG part 2)	Policy answers quote the exact clause — or decline
	S10	LangGraph — turning the AI into an explicit, auditable workflow	Enforced process: triage → route → resolve
6	S11	Pause for people; memory that lasts — approvals & customer profiles	Approvals resumable days later; customers remembered across tickets
	S12	A team of AI specialists — multi-agent architecture	Supervisor delegating to order / policy / refund experts
7	S13	FastAPI backend — serving the assistant to real users	Web API with streaming chat & approval endpoints
	S14	From product spec to tasks — PRD, acceptance criteria, task list; setting up an AI coding agent on the repo	A spec an engineer or an AI pair can execute without guessing
8	S15	A modern web interface — built with AI pair-programming (Claude Code)	Full web app: chat, approval screen, report page
	S16	Cloud deployment & automation — going live, scheduled daily reports	Running on the internet with durable memory
9	S17	Fine-tuning I — when to fine-tune; a training set from real tickets; LoRA/QLoRA on a free GPU	A small open-weights model trained on the shop's own tickets
	S18	Fine-tuning II — honest evaluation, serving your own model, deciding what ships	Ticket triage on a self-hosted model, with a measured fallback
10	S19	Agentic engineering — coding-agent skills, subagents, hooks, MCP; the assistant's tools as an MCP server	A codebase an AI agent can work on safely — and query
	S20	Final presentations — students demo their own extensions; evaluation	One personalized capstone per student, defended live

Training outcomes

On completion, participants are able to:

- ✓ Design and implement AI/LLM applications end-to-end — agents, retrieval, memory, safety, and human oversight
- ✓ Serve and deploy an AI system as a real product (API, web UI, cloud)
- ✓ Debug AI behavior systematically from execution traces
- ✓ Apply industry engineering discipline: versioning, testing, cost control, observability
- ✓ Fine-tune, evaluate and serve a small open-weights model; direct AI coding agents with specs, review gates and MCP
- ✓ Present and defend a working system — each student demos a personalized capstone

Assessment & materials

- ✓ Graded exercises with solutions in every session
- ✓ Continuous project milestones (20 committed versions)
- ✓ Rubric-based capstone evaluation with a final demo day
- ✓ All materials provided: runnable notebooks, project repository, an advanced self-study track, and an interview-preparation question bank
- ✓ Certificate of completion issued by Grow with Data

Prerequisites

- ▶ Programming fundamentals & basic Python
- ▶ A beginner's familiarity with AI/LLMs (having used ChatGPT is sufficient)
- ▶ Own laptop — all tools and AI models used are free-tier; no cost to students

Conducted by Grow with Data

Grow with Data delivers industry-grade, hands-on AI engineering training, designed and taught by practitioners who ship AI products in production for the US market.

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Program coordination & scheduling

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